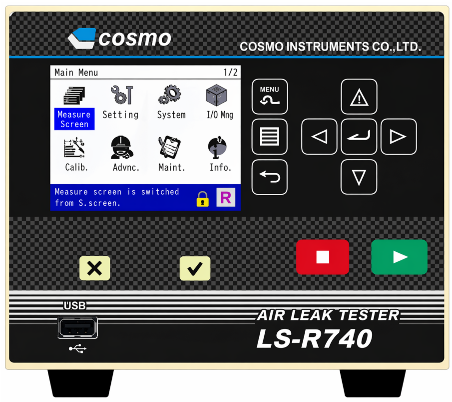
COSMO INSTRUMENTS (I) PVT. LTD.

SPECIALIST IN AIR LEAK AND FLOW TEST



AIR LEAK TESTER

MODEL LS-R740

ERROR AND TROUBLESHOOTING MANUAL

1 Error

1.1 When an Error Occurred

The error code is displayed when an error occurs.

Check the description of the error in Error List.

1.2 Error List

Menu to view descriptions, probable causes and treatments and all the errors.

Go to: Main Menu > Troubleshooting > Error List

The errors are divided into every 9 codes.

Move the cursor to the Error code of your choice and press Enter.

The probable cause and its treatment will be displayed.

2 Error Messages and Treatments

NOTE In LS-R740ZL, BLW stage is not carried out. Errors in Large Leak DET stage occur only in the periods shaded in gray in the timing charts. In addition, the periods shaded with diagonal lines indicate that errors occur in Small Leak DET stage.

2.1 ERROR 1: PS Offset Error

Timing: During power-on check procedure

Criteria: Pressure sensor (PS) offset exceeds $\pm 2\%$ of its range.

Probable Cause	Treatment
Test pressure sensor (PS) offset is out of tolerance when the power is turned on.	Adjust the PS offset. Go to: Maint. > Inspection > Sensor > PS Contact Cosmo for repair if the offset exceeds $\pm 2\%$ of the sensor range.

2.2 ERROR 2 PS Out of Range

Timing: Always monitored during Large Leak CHG (L-CHG) and Small Leak CHG (S-CHG) stage

Criteria: Test pressure exceeds the sensor range.

Probable Cause	Treatment
Test pressure sensor (PS) was pressurized exceeding the sensor full-scale.	Adjust the test pressure. Pay extra attention for low pressure models.
Test pressure sensor (PS) offset is out of tolerance.	Adjust the PS offset. Go to: Maint. > Inspection > Sensor > PS Contact Cosmo for repair if the offset exceeds $\pm 2\%$ of the sensor range.
Cable disconnection or malfunction of the test pressure sensor (PS)	Contact Cosmo for repair.

2.3 ERROR 3 Improper Test Pressure

Timing: Test pressure too low: At the end of Large Leak CHG (L-CHG) and Small Leak CHG (S-CHG) stage
Test pressure too high: Always monitored

Criteria: Test pressure exceeds upper or lower limit.

Probable Cause	Treatment
Zero "0" was set to the Lower Limit.	Set other numerical value than "0" to the Lower Limit.
Upper and Lower limits for Test pressure are too close or inappropriate.	Set appropriate Upper and Lower limits for Test pressure. Go to: Settings > Advanced Settings > Test Press > Upper Press Limit (L-CHG) / Lower Press Limit (L-CHG) / Upper Press Limit (S-CHG) / Lower Press Limit (S-CHG)
Pressurization time is insufficient. (When pressure is lowered.)	Extend Large Leak CHG (L-CHG) timer and/or Small Leak CHG (S-CHG) timer. Go to: Settings > Advanced Settings > Timer > Large Leak CHG (L-CHG) / Small Leak CHG (S-CHG)
Fluctuation or a drop in the source pressure	Check the source pressure or the regulator setting. Avoid using air tools branching off the pressure source of the LS-R740 to supply a stable air. Setting up a dedicated pressure source for the LS-R740 is recommended.
Leaks from seals, part and fittings	Check the seals, part and fittings for possible leaks.
Malfunction of the test pressure sensor (PS)	Contact Cosmo for repair.

2.4 ERROR 10: DPS Offset Error

Timing: During power-on check procedure

Criteria: Differential Pressure sensor (DPS) offset exceeds $\pm 30\%$ of its range.

Probable Cause	Treatment
Differential pressure sensor (DPS) offset is out of range when the power is turned on.	Adjust the DPS offset. Go to: Maint. > Inspection > Sensor > DPS Contact Cosmo for repair if the offset exceeds $\pm 30\%$.

2.5 ERROR 12 Air Operated Valve Error 2

Timing: At the end of Large Leak CHG (L-CHG) and Small Leak CHG (S-CHG) stage

Criteria: Auto-zero of PS is smaller than 1% of the sensor range at the end of L-CHG or S-CHG stage

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the LS-R740. Setting up a dedicated pressure source for the LS-R740 is recommended.
Pressure source is disconnected.	Check the pressure source and the regulator setting.
Test pressure is too low for high pressure model (H20).	Adjust the test pressure within the test pressure range.
Malfunction of the test pressure sensor (PS), solenoid valve or air-operated valve.	Contact Cosmo for repair.

2.6 ERROR 14 Air Operated Valve Error 4

Applicable only to LS-R740WL.

Timing: At the end of Air-Blow (BLW) stage

Criteria: The differential pressure during Air-Blow did not reach the Blow ΔP Limit.

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the LS-R740. Setting up a dedicated pressure source for LS-R740 is recommended.
Air-Blow (BLW) timer is too short or Blow ΔP Limit is too high.	Extend the Air-Blow (BLW) timer or lower the Blow ΔP Limit. Air-Blow (BLW) Timer: Go to: Settings > Advanced Settings > Timer > Air-Blow (BLW) Blow ΔP Limit: Go to: Settings > Advanced Settings > Self Check > Blow ΔP Limit
Malfunction of the test pressure sensor (PS), solenoid valve or air-operated valve.	Contact Cosmo for repair.

2.7 ERROR 15: Air Operated Valve Error 5

Timing: At the end of Large Leak BAL (L-BAL)

Criteria: Pressure switch monitoring the pilot pressure for the Balance (BAL) valve is not activated.

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the LS-R740. Setting up a dedicated pressure source for LS-R740 is recommended.
Malfunction of the pressure switch monitoring the pilot pressure for the Balance (BAL) valve.	Contact Cosmo for repair. As a provisional measure, the pressure switch monitoring can be disabled.

	Go to: Settings > Common Settings > Special > PSW Monitoring > Disable

2.8 ERROR 16: Air Operated Valve Error 6

Timing: During idle state

Criteria: DPS offset exceeded the Idle ΔP Check limit within the programmed Idle ΔP Check Time

Probable Cause	Treatment
DPS offset exceeded its monitoring limit while the LS-R740 is in idle state.	Adjust DPS offset. Go to: Maint. > Inspection > Sensor > DPS Contact Cosmo for repair if the DPS offset exceeds $\pm 30\%$ of its range.
Exhaust time is insufficient.	Extend Idle ΔP Check Timer or Exhaust timer. Idle ΔP Check Timer: Go to: Settings > Advanced Settings > Self Check > Idle ΔP Check Timer Exhaust timer: Go to: Settings > Advanced Settings > Timer > Exhaust (EXH)
Malfunction of the fill valve: SV1 or AV1	Contact Cosmo for repair.

2.9 ERROR 21 DPS Stopped Oscillating

Timing: Always monitored

Criteria: DPS stopped oscillating.

Probable Cause	Treatment
Malfunction of the DPS or power source or cable disconnection	Contact Cosmo for repair.

2.10 ERROR 24 K(Ve) Value Out of Range

Timing: At the end of last DET in K(Ve) Automatic Setup

Criteria: Calculated K(Ve) exceeded 100L.

Probable Cause	Treatment
The current K(Ve) settings do not match the calibrator used for K(Ve) Automatic Setup causing the measured value exceeding 100L.	Check the settings for the calibrator. Go to: K(Ve) > K(Ve) Settings > Basic The items to be set vary depending on the calibrator used for K(Ve) Automatic Setup. ALC: LC Displacement or LC Reading Leak Master: LM Flow [mL/min]

2.11 ERROR 25: Leak Limit Out of Range

Timing: At the end of last DET in K(Ve) Automatic Setup

Criteria: K(Ve) Leak limits exceeded the DPS range after K(Ve) Automatic Setup

Probable Cause	Treatment
Leak limits exceeded the DPS range after executing the K(Ve) Automatic Setup.	Change the Leak unit to a pressure unit and perform K(Ve) Automatic Setup again. Go to: Settings > Advanced Settings > Unit > Leak

2.12 ERROR 30 Large Leak Comp Error

Timing: At the end of Small Leak DET (S-DET) stage

Criteria: The DPS exceeded the Comp Upper Limit or Comp Lower Limit at the end of Large Leak DET (L-DET).

Probable Cause	Treatment
Large difference between Work volume and Master volume.	Reduce the difference between Work volume and Master volume.
Improper ALC displacement	Check whether the ALC calibrations are correct.
Improper large leak comp limit is set.	Adjust the large leak comp limit. Go to: Comp > Large Leak DET Comp > Comp Upper Limit / Comp Lower Limit

2.13 ERROR 31: Small Leak Comp Error

Timing: At the end of Small Leak DET (S-DET) stage

Criteria: The DPS exceeded the Comp Upper Limit or Comp Lower Limit at the end of Small Leak DET (S-DET).

Probable Cause	Treatment
There is a leak in tester circuit or Work.	Check for the leak.
Pressurization and stabilization time is insufficient.	Set longer L-CHG, L-BAL, and S-CHG, S-BAL timers. Go to: Settings > Advanced Settings > Timer > Large Leak CHG (L-CHG) / Large Leak BAL (L-BAL) / Small Leak CHG (S-CHG) / Small Leak BAL (S-BAL) Note: S-CHG and S-BAL timers are set only in LS-R740WL.
Improper small leak comp limit is set.	Adjust the small leak comp limit. Go to: Comp > Small Leak DET Comp > Comp Upper Limit / Comp Lower Limit

2.14 ERROR 32 ALC Error

Timing: At the end of Large Leak DET (L-DET) stage

Criteria: The DPS did not reach the value set for checking ALC operation.

Probable Cause	Treatment
Pilot pressure drop	Adjust the pilot pressure.
Improper ALC calibrations	Check whether the ALC calibrations are correct.

Improper ALC operation check value	Adjust the ALC operation check value. Go to: Settings > Advanced Settings > Self Check > ALC Operation Check
ALC malfunction	In case of ALC or solenoid valve malfunction, contact Cosmo for repair.

2.15 ERROR 33 DPS Sensitivity Error

Timing: At the end of Large Leak DET (L-DET) stage

Criteria: The DPS did not reach the value set for checking AV operation.

Probable Cause	Treatment
Pilot pressure drop	Adjust the pilot pressure.
Improper setting of judgment value for AV operation check.	Correct the value set for AV operation check. Go to: Settings > Advanced Settings > Self Check > AV Operation Check
Malfunction of ALC, air-operated valve, or DPS	In case of ALC, air-operated valve or DPS malfunction, contact Cosmo for repair.

2.16 ERROR 34 DPS Sensitivity Error (EXH)

Timing: At the end of Exhaust (EXH) stage

Criteria: The differential pressure during Exhaust (EXH) stage did not reach the value set for AV Operation Check (EXH).

Probable Cause	Treatment
Improper setting of judgment value for AV operation check (EXH)	Adjust the value set for AV operation check (EXH). Go to: Settings > Advanced Settings > Self Check > AV Operation Check (EXH)
Air-operated valve or DPS malfunction	In case of air-operated valve or DPS malfunction, contact Cosmo for repair.

2.17 ERROR 51 to ERROR 61: System Errors

Usually system errors (ERROR 51 to ERROR 61) are caused by malfunction of electrical components.

NOTE: ERROR 51 (Lo Battery SRAM Error) may occur because of the battery life.

System Errors (Judgment timing: At the time measurement is activated)

Error Code	Error Message	Description
ERROR 51	Lo Battery SRAM Error	Probable Cause 1: The battery has discharged completely. Treatment 1: Refer to the procedure in the manual (8 Maintenance) and replace the battery immediately. Probable Cause 2: If the same error occurs after replacing the battery, some electrical parts may be malfunctioned.

		Treatment 2: Contact Cosmo for repair after executing a System Backup.
ERROR 52	SPI2_res	AD Communication failure
ERROR 53	SPI1-res	I/O Communication failure
ERROR 59	Flash data area bad track Error	E2PROM error
ERROR 60	Flash program area WR Error Kernel	SD card error
ERROR 61	RAM checksum Error	SRAM checksum error

Contact Cosmo for repair after executing System Backup.

3 Large Leak List

The cause of error depends on in which stage Large Leak judgment is made.

Display	Probable Cause	Treatment
CHG Large Leak UL CHG Large Leak LL	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
DL2 Large Leak UL DL2 Large Leak LL	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Small Leak CHG (S-CHG) timer. Go to: Settings > Advanced Settings > Timer > Small Leak CHG (S-CHG)
BAL2 Large Leak UL BAL2 Large Leak LL	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Small Leak CHG (S-CHG) timer. Go to: Settings > Advanced Settings > Timer > Small Leak CHG (S-CHG)
DET Large Leak UL DET Large Leak LL	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Small Leak CHG (S-CHG) timer and Small Leak BAL (S-BAL) timer. Go to: Settings > Advanced Settings > Timer > Small Leak CHG (S-CHG) / Small Leak BAL (S-BAL)
L-BAL Large Leak UL L-BAL Large Leak LL	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Large Leak CHG (L-CHG) timer. Go to: Settings > Advanced Settings > Timer > Large Leak CHG (L-CHG)
L-DET Large Leak UL L-DET Large Leak LL	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Large Leak CHG (L-CHG) timer and Large Leak BAL (L-BAL) timer.

		Go to: Settings > Advanced Settings > Timer > Large Leak CHG (L-CHG) / Large Leak BAL (L-BAL)

If the problem persists without identifiable causes, conduct the No Leak Check.

1) Close both the WORK and MASTER stop valves on the rear panel of the tester.

2) Go to: Main Menu > Maint. > Inspection > Leak Check > No-Leak Check

Contact Cosmo for repair, if internal leak is found.

4 Frequent (+) Fails

Follow the procedures below to identify the cause of the frequent fails and remedy the problem.

1 Perform a No-Leak Check with the stop valves on the rear panel closed.

If the LS-R740 is not leaking, the cause of the frequent fails is something else. Proceed to the next item to check. Contact Cosmo for repair if internal leak is found.

2 Check the fixture condition.

Probable Cause	Treatment
Leaks from tube fittings	Look for leaks in the fittings by performing a bubble test applying soap solution. Redo the tubing if needed.
Deformation of tube	Replace the tube with the harder one that does not deform with the air pressure.

Proceed to the next item to check if the problem persists without identifiable causes.

3 Check the sealing condition.

Probable Cause	Treatment
Sealing material is missing.	Place the sealing material.
Sealing surface is contaminated.	Clean the sealing surface.
Sealing material is damaged or worn-out.	Replace it with a new one.
Sealing deforms when the fixture clamps.	Check the following: – Whether the clearance between the sealing material and the groove is enough. – Wear of the stopper. – Whether the size and hardness of the sealing material are appropriate. – If the thrust force of the cylinder has lowered.

Proceed to the next item to check if the problem persists without identifiable causes.

4 Check whether there were environmental changes.

Probable Cause	Treatment
Tested part is exposed to the direct wind from air conditioner or fan.	Move the source of the wind to where the wind does not directly hit the tested parts.
Some air tools are branched off the pressure source for the LS-R740 causing fluctuation in the pressure source.	Avoid using air tools branching off the pressure source for the tester to supply a stable air. Setting up a dedicated pressure source for the LS-R740 is recommended.
Air compressor capacity is insufficient.	Use the air compressor whose capacity is large enough.
The current compensation value may not be suitable for the current environmental condition.	Update the compensation value.

Proceed to the next item to check if the problem persists without identifiable causes.

5 Check the condition of the tested parts.

Probable Cause	Treatment
Part temperature is higher or lower than the ambient temperature.	Let the part temperature be ambient by adding a cooling/warming buffer in the production line.
The tested parts are wet.	Improve the drying process or add one if there isn't any.
The tested parts get deformed by pressurization.	Add a stopper to prevent the deformation.
Leak due to the gas porosity or internal leak	Look for leaks in the fittings by performing a bubble test applying soap solution. Redo the tubing if needed. If no leak is confirmed, there may be internal leaks. If there is a leak, re-evaluate the production process.

5 Frequent (-) Fails

There are two types of causes for the negative fails. One is caused by a pressure rise in WORK-side circuit and the other is caused by a pressure reduction in the MASTER side circuit.

Follow the procedures below to identify the cause of the frequent fails and remedy the problem.

1 Perform a No-Leak Check with the stop valves on the rear panel closed.

If the LS-R740 is not leaking, the cause of the frequent fails is something else. Proceed to the next item to check. Contact Cosmo for repair if internal leak is found.

2 By a pressure rise in WORK-side circuit

Probable Cause	Treatment
Sealing is not stable.	Check the following: – Whether there is enough clearance between the sealing material and the groove. – Wear of the stopper. – Whether the size and hardness of the sealing material are appropriate. – Whether the thrust force of the cylinder is too high.
A rise in temperature of the air inside the tested part due to the temperature rise of the cold tested part trying to match the ambient temperature.	Let the part temperature be ambient by adding a cooling/warming buffer in the production line. If the part is wet, add or improve the drying process.

Proceed to the next item to check if the problem persists without identifiable causes.

3 By a pressure reduction in MASTER side circuit

Probable Cause	Treatment
There are leaks from the Master or the fittings on the MASTER side.	Check Master part and the fittings for possible leaks. Replace the Master part and fittings if leaks are found.
Deformation of MASTER side tube	Replace it with tube that is rigid enough not to deform.
Adiabatic compression effect of the Master	The size of the Master Chamber may be wrong, or the BAL2 timer may be too short. Replace the master to the one with good temperature stability. Extend BAL2 timer if possible.

Proceed to the next item to check if the problem persists without identifiable causes.

4 By over compensation

Probable Cause	Treatment
The current compensation value may not be suitable for the current environmental condition.	Update the compensation value.